

Set 1

Standard: 6

Subject: Science

Time: 1 Hour

Max Marks: 20

Chapter 16: The Universe

Q.1. Objectives (4 Marks)

- (a) The birth place of stars are huge _____, made of dust particles and gases.
- (b) The planet _____ appears as if it is rolling along its orbit.
- (c) State whether True or False: The density of the planet Saturn is very high.
- (d) Find the odd man out: Spiral, Elliptical, Regular, Irregular.

Q.2. Short Answer (Pick 5, Solve Any 4) (8 Marks)

- (a) What is the difference between stars and planets?
- (b) What is to be found between Mars and Jupiter?
- (c) What are the types of galaxies?
- (d) Name the different types of stars.
- (e) Name the planets without a single satellite.

Q.3. Brief/Long Answer (Pick 3, Solve Any 2) (8 Marks)

- (a) What are the types of comets and on what basis are they classified?
- (b) What are the characteristics of the planet Neptune?
- (c) What is a special characteristic of the planet Mars?

Set 2

Standard: 6

Subject: Science

Time: 1 Hour

Max Marks: 20

Chapter 16: The Universe

Q.1. Objectives (4 Marks)

- (a) Our solar system is located in the _____ galaxy.
- (b) The planet _____ is also known as the "Stormy Planet".
- (c) State whether True or False: The Moon has a very thick atmosphere.
- (d) Find the odd man out: Sirius, Earth, Alpha Centauri, Polaris.

Q.2. Short Answer (Pick 5, Solve Any 4) (8 Marks)

- (a) Which celestial bodies form the solar system?
- (b) Which galaxy is our closest neighbour?
- (c) Name the planet with the largest number of satellites.
- (d) What is the surface temperature of the Sun?
- (e) What is a dwarf planet? Give an example.

Q.3. Brief/Long Answer (Pick 3, Solve Any 2) (8 Marks)

- (a) Which celestial bodies does a galaxy include?
- (b) What is the difference between meteors and meteorites?
- (c) Write a short note on the characteristics of Venus.

ANSWER KEYS

Set 1

Q.1.

- (a) nebulae
- (b) Uranus
- (c) False
- (d) Regular

Q.2.

(a) Difference:

- Stars are luminous and emit their own light.
- Planets are non-luminous and reflect light from stars.
- Stars are stationary relative to each other in a galaxy.
- Planets revolve around a star in fixed orbits.

(b) Between Mars and Jupiter:

- There is a belt of small-sized bodies.
- These bodies are known as asteroids.
- They could not turn into planets when the solar system was formed.
- They continue to revolve independently around the Sun.

(c) Types of galaxies:

- Spiral galaxies.
- Elliptical galaxies.
- Barred spiral galaxies.
- Irregular galaxies.

(d) Types of stars:

- Sun-like stars.
- Red Giants.
- Super Nova.

- Binary or Twin Stars.

(e) Planets without satellites:

- Mercury does not have any satellites.
- Venus does not have any satellites.
- These are the only two planets in the solar system without satellites.
- They are also the planets closest to the Sun.

Q.3.

(a) Types of Comets:

- Comets are classified into two main groups based on their period of revolution.
- Long period comets: These take more than 200 years to complete one revolution around the Sun.
- Short period comets: These take less than 200 years to complete one revolution around the Sun.
- Comets are made of frozen matter and dust particles.
- When close to the Sun, frozen matter converts to gas due to solar heat.
- These gases are thrown away from the Sun, forming a long feathery tail.

(b) Characteristics of Neptune:

- A single season on Neptune lasts for about 41 years.
- On this planet, winds blow with extremely high speeds.
- It is classified as an outer planet.
- It has 13 known satellites.
- It possesses a gaseous outer cover and rings.

(c) Characteristics of Mars:

- The soil on Mars contains a significant amount of iron.
- Due to the iron content, the planet appears reddish in colour.
- Because of its colour, it is also known as the "Red Planet".
- It has the highest and longest mountain in the solar system called 'Olympus Mons'.
- It has two known satellites.

Set 2

Q.1.

- (a) Milky Way (Mandakini)
- (b) Jupiter
- (c) False
- (d) Earth

Q.2.

(a) Solar System components:

- The Sun, which is at the centre.
- Eight major planets.
- Asteroids and dwarf planets.
- Comets and meteors.

(b) Closest Neighbour:

- The galaxy closest to our Milky Way is Andromeda.
- It is a part of the Local Group of galaxies.
- It is visible as a faint patch in the sky.
- It is millions of light-years away.

(c) Planet with most satellites:

- Jupiter has the largest number of satellites.
- It has 64 known satellites.
- Its huge gravitational pull allows it to hold many moons.
- It is the largest planet in the solar system.

(d) Sun's temperature:

- The Sun is a yellow-coloured star.
- Its surface temperature is around 6000°C.
- It provides heat and light energy to the entire solar system.
- The temperature at its core is much higher than the surface.

(e) Dwarf Planet:

- It is a small-sized celestial body.
- It revolves independently around the Sun.
- Pluto is the best-known example of a dwarf planet.
- Pluto takes around 248 years to complete one revolution.

Q.3.

(a) Celestial bodies in a galaxy:

- Innumerable stars of various sizes.
- Diverse planetary systems associated with stars.
- Large clusters of stars.
- Huge nebulae made of dust and gases.
- Clouds of gases and dust particles.
- Remnants of dead stars and newly born stars.

(b) Meteors and Meteorites:

- A meteor is a small piece of stone or metal entering the atmosphere.
- It is commonly called a "falling star" as it burns up due to friction.
- Most meteors get completely burnt in the atmosphere.
- If a meteor is large and does not burn completely, it reaches the earth's surface.
- The unburnt part that falls on the ground is called a meteorite.
- Meteorites can cause large craters on the surface of planets like Mercury.

(c) Characteristics of Venus:

- Venus is the second planet from the Sun.
- it is the brightest planet in our solar system.
- It is also the hottest planet due to its thick atmosphere.
- It is visible in the east before sunrise and in the west after sunset.
- Unlike most planets, it rotates around itself from east to west.
- It does not have any natural satellites.